

SIDDHARTH GHODE

Backend / Full-Stack Software Engineer

9307899172 | siddharthghode59911@gmail.com | Pune, India

[LinkedIn](#) [GitHub](#)

EDUCATION

Master of Computer Science Pune University, Department of Computer Science	2024–Present
Bachelor of Computer Science Savitribai Phule Pune University	2021–2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C, C++
Backend: Django, Django REST Framework, REST APIs, JWT, RBAC, OAuth 2.0
Frontend: React 19, Tailwind CSS, HTML5, CSS3, Bootstrap
Databases: PostgreSQL, MySQL, SQLite, MongoDB
Tools: Git, GitHub, Docker, Linux, VS Code
Data / ML: Pandas, NumPy, Scikit-learn, TF-IDF, Cosine Similarity
System Design: Caching, Load Balancing, Rate Limiting, CDN, Message Queues

PROJECTS

Library Management System – Intelligent Book Recommendation Platform

Python • Django REST Framework • React • PostgreSQL

[GitHub](#)

- Architected 25+ REST APIs across authentication, catalog management, borrowing workflows, analytics, messaging, and recommendation services using a modular Service-Layer architecture with Django REST Framework.
- Engineered a hybrid recommendation engine combining TF-IDF, Cosine Similarity, Collaborative Filtering, and Truncated SVD with multi-tier fallback strategies to deliver personalized recommendations across a 6,000+ book catalog.
- Designed a secure authentication and authorization pipeline using JWT, Google OAuth 2.0, Argon2 password hashing, and department-scoped RBAC, enabling stateless multi-tenant access control.
- Implemented atomic borrow transactions using `transaction.atomic()` and `select_for_update()`, eliminating race conditions and ensuring consistent inventory updates under concurrent requests.
- Optimized recommendation and analytics workloads using Django Cache, `select_related()`, bulk database operations, and database-side weighted scoring, reducing repeated computation for frequently accessed queries.

PortfolioHub – Full Stack Portfolio Management Platform

Python • Django REST Framework • React • PostgreSQL

[GitHub](#)

- Developed a portfolio management platform enabling authenticated users to create, edit, organize, and share projects through secure REST APIs built with Django REST Framework and PostgreSQL.
- Implemented secure CRUD operations, media uploads, pagination, filtering, search, JWT authentication, OAuth 2.0, session management, and device tracking with Docker-based deployment.
- Designed a modular frontend using React 19 with protected routes, reusable components, Axios-based API integration, and responsive UI built with Tailwind CSS.

Data Processing & Analytics Pipelines

Python • Pandas • NumPy

- Built reusable analytics pipelines for CSV, Excel, and JSON datasets using filtering, aggregation, joins, and validation.

CERTIFICATIONS

Data Visualisation

[Certificate](#)

SUMMARY

Backend / Full-Stack engineer specializing in Python, Django REST Framework, React, PostgreSQL, and Java Spring Boot. Built production-style applications featuring REST APIs, JWT authentication, RBAC, scalable database design, Docker, and ML-powered recommendation systems. Passionate about backend architecture, clean code, Linux, and distributed systems.